

Design and Development of a Multi-Sensor Smart Bandage for Real-Time Wound Condition Monitoring and Infection Detection

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**RESEARCH METHODOLOGY AND PROJECT
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1. ABSTRACT

Wound management is a major global healthcare challenge, particularly in patients suffering from different types of wounds. Traditional wound care methods rely heavily on visual inspection and periodic dressing changes, which can delay infection detection and lead to poor healing outcomes. This research aims to design and develop a multi-sensor smart bandage capable of continuously monitoring wound parameters such as pH, temperature, and pressure, while integrating RGB image analysis for real-time wound assessment. The proposed system will utilize a microcontroller-based data acquisition unit to process and transmit sensor data for evaluation and control. Additionally, the design incorporates a programmable drug delivery system and a Peltier-based cooling mechanism to provide automated wound treatment. Through systematic design, simulation, and prototype development, this study seeks to create a closed-loop wound monitoring and management system that enhances healing efficiency, reduces hospital visits, and minimizes infection risk. The expected outcome is an intelligent, affordable, and portable wound care solution suitable for both clinical and home-based applications.

2. INTRODUCTION

A wound is an injury that breaks the skin or other body tissue. Wounds can be open, with broken skin and exposed body tissue, or closed when there is damage to tissue under intact skin. Every year, wounds have an important economic impact on the health systems of each country since the treatment of chronic wounds depends on wound etiology. Generally, adequate wound cleaning and debridement, control of possible infections, and the use of wound dressings are needed. Nowadays, current therapies cannot fully address impaired healing, thus provoking wound complications like infections and poor wound closure. For these reasons, new therapies that promote adequate wound healing, preventing wound infections and inflammation, are necessary. In this way, the integration of nanomaterials with remarkable properties into wound healing has led to significantly positive results. For a deeper understanding of wound healing advances and challenges, in this chapter, we review the different types of wounds, their mechanism, and pathophysiology. [1] Wounds can be classified in many ways, by acute or chronic, by cause (e.g., pressure, trauma, venous leg ulcer, diabetic foot ulcer), by the depth of tissue involvement, or other characteristics such as closure

* Acute wound

An acute wound is defined as a recent wound that has yet to progress through the sequential stages of wound healing.³ An acute wound is acquired as a result of an incision or trauma and heals in a timely and orderly manner. Surgically created wounds include all incisions, excisions, and wounds that are surgically debrided. Surgical wounds include all skin lesions that occur as a result of trauma (e.g. burns, falls), as a result of an underlying condition (e.g. leg ulcers), or as a combination of both.

* Chronic wounds

Wounds that fail to heal in an anticipated time frame and orderly fashion and often recur are considered chronic.³ Venous leg ulcers, pressure ulcers and diabetic foot ulcers are some examples of chronic wounds. [2]

2.1 Problem statement

Wound management is a critical aspect of modern healthcare, particularly for patients suffering from chronic wounds, post-surgical wounds, diabetic ulcers, pressure sores, and burn injuries. Traditional wound care relies heavily on manual inspection by healthcare professionals to monitor healing progress and detect signs of infection. However, this conventional approach has several limitations, including delayed identification of complications, frequent dressing changes, increased hospital visits, and a higher risk of secondary infections. As the prevalence of chronic wounds continues to rise globally, there is an urgent need for innovative, technology-driven solutions that can provide continuous, accurate, and real-time wound monitoring.

2.2 Aim

To explore, design, and develop a multi-sensor smart bandage system that enables real-time monitoring of wound healing parameters and assists in early detection of infections.

2.3 Research Objectives

1 – Literature & Technology Review

To review and analyze existing technologies, sensing mechanisms, and materials used in smart bandages for wound condition monitoring and infection detection.

(Purpose: Build a strong foundation and identify knowledge gaps.)

2 – Sensor Selection & Feasibility Analysis

To identify, compare, and evaluate potential sensors capable of monitoring wound healing parameters such as pH, temperature, lactate level, moisture, and oxygen concentration, considering factors like sensitivity, accuracy, biocompatibility, power consumption, and integration feasibility.

(Purpose: Choose the most suitable sensing technologies.)

3 – System Design & Integration

To design a multi-sensor smart bandage architecture, integrating selected sensors with flexible materials, microcontrollers, wireless communication modules, and power management systems for real-time data acquisition.

(Purpose: Develop the complete hardware and integration strategy.)

4 – Data Acquisition & Monitoring Framework

To develop algorithms and a monitoring interface for processing, visualizing, and analyzing wound condition data, enabling early detection of potential infections.

(Purpose: Provide actionable insights to patients and healthcare professionals.)

5 – Prototype Development & Validation

3. LITERATURE REVIEW

Chronic wounds (e.g., diabetic foot ulcers, venous leg ulcers, pressure ulcers) affect millions worldwide and drive substantial morbidity and costs. Unlike acute wounds that progress through hemostasis → inflammation → proliferation → remodeling, chronic wounds often stall in inflammation, with dysregulated biochemistry and frequent infection. Key microenvironmental factors—pH, temperature, moisture/exudate, and oxygenation—shape cellular behavior, bacterial growth, and healing trajectories. An acidic milieu generally favors healing, whereas alkaline conditions correlate with biofilm persistence and delayed closure. Local hypoxia (low tissue oxygen) impairs collagen synthesis and angiogenesis, prolonging non-healing states. [23,24]

pH. Acute wounds tend to be near neutral to mildly acidic; chronic wounds commonly shift alkaline. Reviews and in-vivo studies report acute mean pH ≈ 7.44 , with chronic wounds spanning ~ 7.4 – 8.9 (and higher in some diabetic ulcers), linking alkalinity to protease activity, biofilms, and impaired matrix deposition. Lowering pH can reduce bacterial burden and promote healing. [25,26]

Temperature. Local temperature reflects perfusion and inflammation; elevations suggest infection, while depressed temperature can indicate ischemia. Continuous monitoring is emphasized in smart dressing reviews as a non-invasive early signal of complications. [27]

Oxygenation (TcPO₂). Transcutaneous oxygen tension (TcPO₂) is a well-studied predictor of healing potential in limb wounds: values >40 mmHg associate with higher likelihood of healing, whereas <20 – 30 mmHg indicates ischemic risk; some studies debate exact cut-points but agree on the prognostic value. [28,29]

Moisture/exudate. Maintaining balanced moisture supports keratinocyte migration; chronic wounds often show excess exudate (maceration) or dryness (desiccation), each impeding repair. Smart dressings increasingly integrate moisture sensors to maintain the “therapeutic window.” [30]

Recent devices integrate flexible sensors (pH, temperature, strain/pressure, oxygen proxies) and sometimes actuators (drug release, electrostimulation) into conformable dressings for closed-loop care. Examples include ultra-sensitive polyaniline pH meshes within exudate-draining dressings, and multi-sensor hydrogels that monitor pH, glucose, and deformation while providing anti-infection functions. Comprehensive reviews catalog materials, transduction methods, and clinical readiness levels [31,32,33] Deep learning (e.g., U-Net variants) reliably segments wounds and tissue types (granulation/red, slough/yellow, eschar/black), enabling objective area tracking and tissue-composition quantification—key to assessing progress and distinguishing non-healing states. New community datasets (e.g., FUSeg) and evaluations highlight progress and the need for standardized benchmarks and multi-expert annotations due to inter-rater variability [34,35]

Classification. CNNs can differentiate wound categories (e.g., diabetic, venous, infected vs. non-infected) from RGB images by leveraging color/texture and edge regularity; recent studies underscore improved accuracy when models are trained with curated, multi-class data and expert labels [36]

Single-modality assessments are limited: pH or temperature alone may be confounded by dressings, room temperature, or transient inflammation; images alone can miss subsurface hypoxia or biofilm shifts. Reviews and emerging prototypes argue for multimodal fusion—combining continuous biosensor streams (pH, temperature, moisture, pressure/strain, oxygen proxies) with periodic RGB analyses (area change, tissue proportions, edge irregularity)—to improve early infection detection and healing prediction beyond clinician visual assessment alone. [37,38]

Synthesis of the literature suggests the following typical tendencies (to be validated in your cohort):
Acute (healing): near-skin temperature, near-neutral to mildly acidic pH, shrinking area, increasing granulation %, smoothening edges.[39]

Chronic (non-healing): alkaline pH (~7.4–8.9), fluctuating or low temperature in ischemic cases, little area reduction, higher slough %, irregular/undermined edges. [40,41]

Diabetic foot ulcer: often alkaline pH and perfusion deficits variable temperature; chronic course unless revascularized/offloaded. [42]

Venous ulcer: chronic/alkaline milieu with heavy exudate; RGB shows shallow, irregular margins and slough dominance; compression/offloading alters pressure dynamics. [43]

Infected: elevated local temperature, pH drifts alkaline, increased exudate; RGB reveals erythema and pus/green-yellow components; early detection benefits from continuous sensing. [44,45]

Image pipeline: wound ROI detection → tissue segmentation (granulation/slough/eschar) → features (area cm², % tissue classes, boundary irregularity/circularity). Public challenges (e.g., FUSeg) facilitate benchmarking. [46]

Sensor pipeline: calibrated pH & temperature time-series; moisture index; optional TcPO₂ (or surrogates). Fusion/AI: tabular (sensors) + vision (image features) using Gradient Boosting/XGBoost or multimodal deep nets. Explainability (e.g., SHAP) highlights which biomarkers drive decisions; recent smart bandage studies report high predictive accuracy for outcome prediction with interpretable models. [47]

Gaps and challenges

Clinical datasets: scarcity of large, labeled, paired (sensor + image) datasets; need for per-patient splits and multi-center data to avoid bias. [48]

Standardization: lack of unified protocols for color calibration, ruler scaling, and sensor placement/calibration; variable TcPO₂ thresholds across studies. [49,50]

Real-world deployment: integrating flexible electronics, power/battery management, skin safety, and data security into medical-grade bandages requires translational research and regulatory pathways. [51]

4. RESEARCH GAP

Chronic wounds such as diabetic foot ulcers, venous leg ulcers, and pressure injuries impose a substantial global health burden due to prolonged healing, recurrent infection, and high resource utilization. Unlike acute wounds, which typically progress through hemostasis → inflammation → proliferation → remodeling, chronic wounds often stall in the inflammatory phase, hindered by enzymatic dysregulation, persistent inflammation, and microbial biofilms [58,59]

Key microenvironmental biomarkers- temperature, pH, moisture (exudate), and oxygenation critically influence healing outcomes. Chronic or infected wounds commonly shift toward an alkaline pH ($\approx 7-9+$), which correlates with biofilm persistence, while healthy wounds trend more neutral to mildly acidic [60,61]. Elevated local temperature often precedes infection, and hypoxia (e.g., transcutaneous oxygen tension $< 20-30$ mmHg) correlates with poor healing potential [62,63,64].

While smart dressings incorporating single sensors (e.g., only pH or temperature) exist [65,66], robust multi-sensor fusion systems combining pH, temperature, moisture, oxygen, and RGB imaging are rare. Most monitoring devices lack later therapeutic functions [67,68], and closed-loop systems (that both sense and apply cooling or drug delivery in response to detected risk) are notably underdeveloped [69,70]. Additionally, although computer vision (e.g., convolutional neural networks) is used to segment and classify wound types from images [71,72], few models fuse image-derived features with sensor data to enhance diagnostic accuracy [73], and datasets are limited by incomplete multimodal integration [74,75].

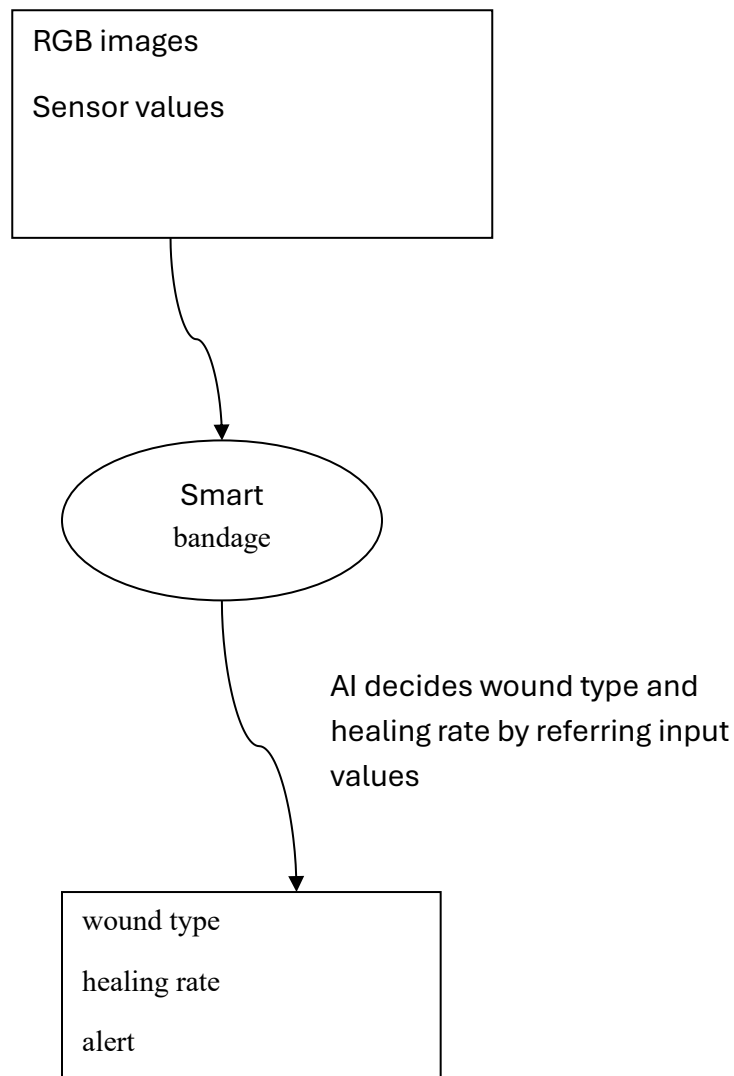
Moreover, practical deployment issues such as energy management, device wearability, thermal safety, dose control, and biocompatibility are commonly overlooked in lab prototypes [76].

This project fills these gaps by developing a multi-sensor smart bandage that fuses temperature, pH, pressure with RGB image features into an AI-driven classifier for wound types (acute, chronic, diabetic ulcer, venous ulcer, infected), and autonomously triggers cooling, heating and controlled drug delivery with safety constraints, enabling a functional closed-loop wound management system.

5. RESEARCH METHODOLOGY

5.1 Workflow/ outline of the methodology

1. RGB images identify wound size, type of edges and wound color.
2. Sensors measure parameter values in wound area.
3. AI analyze RGB image and parameters values and identify wound type by clinical label values.
4. Display wound type and parameter values.
5. After measure parameters temperature continue in stable value with cooling or heating system and PH continue stably with drug release.
6. Measure pressure value continuously and if pressure is higher during 30mins-1hr give alert about that
7. Record wound size within days of clean the wound and guess the healing rate and healing dates and display them.



- sensors use for Multi-Sensor Smart Bandage

temperature sensor – measure temperature

PH sensor – measure PH value for prevent bacterial environment

Pressure sensor – measure pressure value. This is not a blood pressure. It is pressure tape for identify about is wound area swelling (alert when pressure high/swelled)

* Temperature is one of the most commonly recorded physical markers in the body because rates of many enzymatic reactions are temperature dependent. In particular, temperature is related to the inflammation and infection states of wounds [3], and has been adopted as a classic sign of the Clinical Signs and Symptoms Checklist (CSSC) to standardize the assessment of chronic wounds worldwide [4]. The range for the temperature at the skin is from 31.1 °C to 36.5 °C. At the wound site, a prolonged temperature increase of at least 1.11 °C could be due to infections and metabolic activity changes. Specifically, a peri-wound temperature increase of 3 °C to 4 °C has been observed to be correlated with signs and symptoms of infection within 40 participants that have chronic leg ulcers [5]. In the cases with clinically diagnosed wound infection, temperature differential between the wound and healthy skin increased to 4–5 °C, and returned to 0.8–1.1 °C after an adequate treatment with antibiotics [6]. Thus, the peri-wound skin temperature could be used as a useful reference for monitoring the infection symptoms and treatment status.

* pH sensor. A chronic wound bed usually exhibits a pH of 7.15–8.9. However, it is beneficial to have an acidic environment such as pH 4–7 for wound healing since an acidic environment can support fibroblasts proliferation, improve angiogenesis and epithelialization of the wound skin, and prevent bacterial colonization for an open wound [7]. Wearable potentiometry or colorimetric pH sensors could provide important indications on the status of a wound, bacterial infections, and therapeutic management. Guinovart et al. reported a bandage-like, wearable potentiometric sensor for continuous monitoring of wound pH [8].

*Pressure sensing

Real-time monitoring of external pressure exerted on the skin is critical for chronic wound healing, especially diabetic foot ulcers (DFUs) and pressure ulcers. Take DFUs for example, they are a major complication of diabetes, and are hard-to-heal wounds that affect millions of people (2.49 million) in the U.S, with yearly incidence ranging from 2% to 6% [9] and average annual Medicare expenditure of \$29.16 billion [10]. DFUs significantly impact the quality of patients' life as they are slow to heal (78 days on average), frequently recur (40% of DFUs recur within 1 year of healing) and add additional costs. A mainstay of all DFU therapies is mechanical offloading to relieve pressure and stress on the affected foot (usually on the planar surface) [11,12], which can be accomplished through non-weight bearing (like using a walker), the use of an adaptive footwear, or the application of non-removable total contact casting [13]. Flexible pressure sensors have been developed for potential applications in wound monitoring. For example, an inductor-capacitor (LC) wireless pressure sensor was designed and fabricated to monitor the mechanical pressure on the skin wound, as shown in Fig. 3D [14].

RGB images

- * RGB cameras are inexpensive and widely available (from smartphones to medical-grade devices). This makes them practical for clinical and remote monitoring applications. [52]

- * Wound tissues exhibit distinct colors that correspond to physiological states:

 - Red → granulation tissue (healthy healing)

 - Yellow → slough (necrotic/fibrinous tissue)

 - Black → necrosis/eschar

 - Green → infection (e.g., *Pseudomonas* colonization)

- * RGB images capture these spectral differences, enabling classification. [53]

- * Image features such as color histograms, edge irregularity, wound area and texture can be extracted directly from RGB images.

 - * These features can then feed into AI/ML models (CNNs, SVMs) for wound type classification. [54,55]

 - * RGB imaging avoids physical contact with the wound.

 - * Enables longitudinal tracking of healing progression over days/weeks. [56]

- * Combining RGB images with temperature & pH sensors gives a multimodal dataset:

 - RGB → visual wound condition

 - Temperature/pH → microenvironmental status

 - * This strengthens AI-driven recognition of acute, chronic, diabetic, venous, and infected wounds. [57]

Identify wound types by RGB images

Actuate - Red/pink color, smooth edges, minimal slough/necrosis

Chronic - Yellow slough or brown/black necrotic tissue, irregular edges

Diabetic foot - Red/pink color, irregular edges, minimal slough/necrosis

Venous leg ulcer - Yellow slough or brown/black necrotic tissue, irregular edges

Infected wound - Green/yellow pus, irregular edges, presence of pus/slough/necrosis[15]

Parameter values

*Acute temperature: $\sim 34\text{--}37^\circ\text{C}$,

* Chronic temperature: $\sim 30\text{--}33^\circ\text{C}$ [16]

*When the acute wounds occur and the integrity of the skin is compromised, alkaline tissue fluid and plasma in the broken capillaries overflow, the pH in wound increase significantly.¹⁷ The pH of the microenvironment of an acute wound is approximately neutral, with an average value of 7.44. [17]

*chronic diabetic wounds have a relatively alkaline environment, with higher pH values ranging from 7.42 to 8.90. The pH of chronic diabetic wounds can be as high as 9.25. [18]

*foot ulcers at baseline and at 12 weeks. 50 patients (68% [$n = 34$] had diabetes) participated. Mean baseline pH of wounds was 6.95 (SD 1.01); temperature 30.91°C (SD 3.00); and size 0.82 cm^2 . After 12 weeks, 26% ($n = 13$) were lost to follow-up, 50% ($n = 25$) had healed. Of the remaining patients, mean pH was 6.72 (SD 0.54); temperature 30.88°C (SD 2.97), and size 0.13 cm^2 (SD 0.13). [19]

*venous ulcers, often have a more alkaline pH, which can range from 7.15 to 8.9. Temperature variations of on average 0.4°C occurred within venous leg ulcers. [20]

*temperature and if infection occurs in a wound during healing, the temperature will rise above the "inflammation level." In other words, sudden temperature elevation in a wound should raise concerns of infection [21]

*best PH value for wound healing – 5.5 to 6.5

*The cells and enzymes of the body function best at normal temperature, around 37° C (98.6° F). [22]

Inputs and outputs in this system

Inputs – RGB image, temperature value, pressure value , PH value

Outputs – wound type , healing rate, alert

- Apparatus can use for smart bandage

Measure temperature – NTC thermistor / PEDOT: PSS temperature sensor

Measure PH – PH sensor based M-PANI

Pressure tape – flexiforce A201

6. TIME PLAN

- 10 month time plan

Month 1

Collect and review recent papers on smart bandages, wound monitoring (pH, temperature, RGB imaging), drug delivery, and cooling systems. Identify knowledge gaps.

Month 2

Finalize Research Gap & Objectives: Define problem statement, research gap, and detailed objectives. Prepare progress report1

Month 3

Draft conceptual design of smart bandage (sensor integration, RGB camera, microcontroller selection, drug reservoir, cooling mechanism). Draw block diagrams and workflow.

Month 4

Develop detailed methodology – choose materials for bandage, drug delivery method (micropump/controlled release), and cooling system. Conduct simulation/design validation.

Month 5

Prototype Development – Stage 1: Procure sensors (pH, temperature, pressure), RGB camera, microcontroller (ESP32/8051-based), and test them individually.

Month 6

Prototype Development – Stage 2: Integrate multi-sensor system with data acquisition unit. Begin integration of RGB imaging for wound assessment.

Month 7

Prototype Development – Stage 3: Add cooling system (e.g., Peltier-based) and programmable drug delivery mechanism. Conduct initial functionality tests.

Month 8

System Integration & Testing: Integrate all modules (sensors + drug delivery + cooling + RGB camera) into single prototype. Debugging, calibration, and reliability testing.

Month 9 |

Data Collection & Analysis: Simulate wound conditions in lab models (pH, temperature, pressure changes). Record performance of the system (accuracy, response time, control of drug release and cooling).

Month 10

Report Writing & Final Presentation: Analyze experimental results, prepare dissertation, finalize research paper.

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Appendix

*** Meeting outcomes**

He advised me to choose the research gap properly , analyze the topic well and design a good system to suit the research gap.

He gave me knowledge on how to study research papers and find information.

He always helped me to fix the shortcomings in the system and the report.

***discussion time**

Call time duration – 45 mins

Emails- 35+ emails

***Additional**

pressure sensor as a tape. It is not a blood pressure. It is alert about wound area is swelling or not.